FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Main)-2024

TEST DATE: 11-01-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. D

Sol. Since current is leading the source voltage, this shows that circuit is of capacitive nature. However, current leads in phase by $\frac{\pi}{4}$, therefore, box also consists of resistance in series with the capacitance.

$$\because \tan\frac{\pi}{4} = \frac{X_c}{R}$$

$$\Rightarrow X_c = \frac{1}{\omega C} = R \qquad \ldots\ldots\ldots$$

$$I_{max} = \frac{V_{max}}{\sqrt{R^2 + X_c^2}} = \sqrt{2}A \qquad \ldots\ldots\ldots$$

2. A

Sol. The magnetic field at x at the time t is given by

$$B(x,t) = B_0 - 1\frac{T}{m}x - 1\frac{T}{s}t$$

The magnetic flux associated with the loop is given by

$$\Phi_B = \int_{x_1}^{\ell + x_1}\left(B_0 - 1\frac{T}{m}x - 1\frac{T}{s}t\right)\ell dx$$

$$\Rightarrow \Phi_B = B_0\ell^2 - \frac{1}{2}\frac{T}{m}\ell(\ell^2 + 2\ell x_1) - 1\frac{T}{s}t\ell^2$$

$$\because \xi_B = -\frac{d\Phi_B}{dt} = 1\frac{T}{m}\ell^2 v + 1\frac{T}{s}\ell^2 = 0.03\,V$$

$$I = \frac{0.03V}{20} = 1.5\text{ mA}$$

3. A

Sol. $F = B(x+\ell,t)I\ell - B(x,t)I\ell = 1\times 0.1\times 1.5\times 10^{-3}\times 0.1\,N = 1.5\times 10^{-5}\,N$

$\Rightarrow F = 1.5\times 10^{-5}N$

4. C

Sol. The amplitude of the electric field is $E_0 = B_0 c = 10^{-5}\times 3\times 10^8 N/C = 3\times 10^3 N/C$.

The propagation constant $k = \frac{\omega}{c} = \frac{2\pi\times 10^{10}}{3\times 10^8}m^{-1} = \frac{200\pi}{3}m^{-1}$

Since $E(x,t) = E_0 \sin(2\pi\times 10^{10}s^{-1}t - kx)$

$\Rightarrow E(x,t) = 3\times 10^3 N/C \sin(2\pi\times 10^{10}s^{-1}t - \frac{200\pi}{3}m^{-1}x)$

5. B

Sol. $\because\ \hat{r} = \hat{A} - 2(\hat{A}\,.\,\hat{n})\,\hat{n}$.

Here $\hat{k}$ is the normal vector

$\Rightarrow \hat{r} = \hat{A} - 2(\hat{A}\,.\,\hat{k})\,\hat{k}$

$= \hat{A} + \frac{2\sqrt{3}}{2}\hat{k}$

$= \frac{1}{2\sqrt{2}}\,\hat{i} + \frac{1}{2\sqrt{2}}\,\hat{j} + \frac{\sqrt{3}}{2}\hat{k}$

6. B

Sol. Having been projected both, the proton and the alpha particle describe circular motion in x-y plane. Both, proton and the alpha particle move in the clockwise sense. Their angular velocities are $\omega_p = \frac{eB}{m}$ and $\omega_\alpha = \frac{2eB}{4m} = \frac{eB}{2m} = \frac{\omega_p}{2}$ respectively and radii of their corresponding paths are $R_p = \frac{mv}{eB}$ and $R_\alpha = \frac{2mv}{eB}$ respectively. At the time t, velocities of the proton and the alpha particle are given by $\vec{v}_p = v\left[\cos(\omega_p t)\hat{i} - \sin(\omega_p t)\hat{j}\right]$ and $\vec{v}_\alpha = v\left[-\cos(\omega_\alpha t)\hat{i} + \sin(\omega_\alpha t)\hat{j}\right]$ respectively.

When their velocities are mutually perpendicular, $\vec{v}_p\cdot\vec{v}_\alpha = 0$

$\Rightarrow \cos\omega_p t\cos\omega_\alpha t + \sin\omega_p t\sin\omega_\alpha t = 0$

$\Rightarrow \cos(\omega_p - \omega_\alpha)t = 0$ or $\cos\left(\frac{\omega_p}{2}\right)t = 0$

This implies that $\frac{\omega_p}{2}t = \frac{\pi}{2}$ or $\omega_p t = \pi$ and $\omega_\alpha t = \frac{\pi}{2}$,

At this instant coordinate of the proton are, $\frac{mv}{eB}\left[\sin\omega_p t, -\left(1-\cos\omega_p t\right), 0\right] = \left(0, -\frac{2mv}{eB}, 0\right)$ and coordinates of the alpha particle are, $\frac{2mv}{eB}\left[-\sin\omega_\alpha t, \left(1-\cos\omega_\alpha t\right), 0\right] = \left(-\frac{2mv}{eB}, \frac{2mv}{eB}, 0\right)$

Therefore, distance between the is $S=\sqrt{\left(-\frac{2mv}{eB}\right)^2+\left(\left(\frac{4mv}{eB}\right)^2\right)}=\frac{2\sqrt{5}\cdot mv}{eB}=25mm$

7. A
Sol. After 10s, 25% becomes 12.5% $\Rightarrow$ t_{half} = 10 = ln 2/λ
or t_{mean} = 1/λ = 10/ln 2 = 14.43 s.

8. C
Sol. After 10s, 25% becomes 12.5% $\Rightarrow$ t_{half} = 10 = ln 2/λ
Further reduction to 6.25 % of reduced number means becoming $(½)^4$times.
It will take the time of 4 half life times = 10 $\times$ 4 = 40s

9. D
Sol. Use Basic concept

10. C
Sol. Intensity of the transmitted light is given by
$I=I_{max}\cos^2\theta$, where θ is the angle between transmission axis and the direction of polarization.

$\because \frac{3}{4}I_{max}=I_{max}\cos^2\theta$

$\Rightarrow \theta=\frac{\pi}{6},\frac{5\pi}{6},\frac{7\pi}{6}$

However, these angles are achieved when polaroid rotates through an angle 20^0 and 80^0. Therefore, initial angle is 130^0.

11. B
Sol. Hint; friction provides both, centripetal as well as tangential acceleration.
when coin is about to slip on the disc,

$m\sqrt{\left(\omega^2 r\right)^2+\alpha^2 r^2}=\mu mg$

On solving we get
T = 2s.

12. C
Sol. formula, $v=\sqrt{\frac{T}{\mu}}$

Let T_0 be the tension in the string near the axis.

$\Rightarrow T_0=\mu\ell\times\omega^2\times\frac{\ell}{2}$

If tension at the mid-point of the string is T

$T_0-T=\frac{\mu\ell}{2}\times\omega^2\times\frac{\ell}{4}$

$\Rightarrow T=\frac{3\mu\ell^2\omega^2}{8}$

$v=\sqrt{\frac{3}{8}}\omega\ell$

13. C

Sol. formula, $\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mqV}}$

$\because \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha q_\alpha}{m_p q_p}} = 2\sqrt{2}$

14. D

Sol. $y = B\{\sin\omega t\cos\Phi + \cos\omega t\sin\Phi\}$

$\Rightarrow \frac{y}{B} = \frac{x}{A}\cos\Phi + \sqrt{1-\frac{x^2}{A^2}}\sin\Phi$

$\Rightarrow \frac{x^2}{A^2} + \frac{y^2}{B^2} + 2\frac{x}{A}\cdot\frac{y}{B}\cos\Phi = \sin^2\Phi$

For $\Phi = 0$ and π

$\left(\frac{x}{A} \pm \frac{y}{B}\right)^2 = 0$

15. B

Sol. Formula $R = \frac{V^2}{P}$

$\because \frac{P_A}{P_B} = \frac{V^2/R_A}{V^2/R_B} = \frac{5}{4}$

16. A

Sol. formula, $I = I_{max}\cos^2\delta/2$

$\frac{3}{4}I_{max} = I_{max}\cos^2\delta/2$

$\Rightarrow \delta = \frac{\pi}{3}$

If P is at a distance x from the central maximum.

$\frac{2\pi d}{\lambda D}x = \frac{\pi}{3}$

For the second wavelength,

$\delta' = \frac{2\pi d}{\lambda' D}x = \frac{2\pi d}{\lambda' D} \times \frac{\pi}{3}\frac{D\lambda}{2\pi d} = \frac{\pi}{2}$

$I = I_{max}\cos^2\delta'/2 = \frac{I_{max}}{2}$

17. A

Sol. Fission of a nucleus is feasible only if the binding energy of daughter nuclei is more than the parent nucleus.

A = 55 will have more BE than 110.

A = 70 will have same BE as 110 but A = 40 will have more B.E.

A = 100 will have same BE as 110 but A = 10 will have lesser B.E.

A = 90 will have same BE as 110 but A = 20 will have lesser B.E.

18. A
Sol. Use basic concept

19. C
Sol. Use basic concept

20. B
Sol. For potential energy to be zero, U(x) = 0,

$\Rightarrow\ ax^4 - bx^2 = 0$

$\Rightarrow\ x = 0,\ \pm\sqrt{\frac{b}{a}} = \pm 2\sqrt{2}\,m$.

For potential energy to be the maximum or minimum, $\frac{dU}{dx} = 0$.

$\Rightarrow\ 4\,ax^3 - 2bx = 0$

$\Rightarrow\ x = 0,\ \pm 2m$

$\because\ U(x = 0) = 0$

and $U(x = \pm 2) = -16\,J$.

This implies that at x = ± 2m, potential energy is the minimum.
Since total energy is conserved, therefore where potential energy is the minimum, kinetic energy is the maximum. As we know that, initially, both, potential energy and kinetic energy is zero. Therefore, total energy will remain zero.

$\Rightarrow\ U_{min} + K_{max} = 0$

$\Rightarrow\ -16\,J + K_{max} = 0$

$\Rightarrow\ K_{max} = 16\,J$

$\Rightarrow\ v = 4$ m/s

SECTION – B

21. 13
Sol. $(Mg + T)\sin\theta = Ma$ …(i)

$T + ma\sin\theta - mg = 0$ …(ii)

From (i) and (ii)

$$a = \frac{(M+m)g\sin\theta}{M + m\sin^2\theta}$$

22. 2
Sol. $\because \frac{E_0Z^2}{n^2} + \frac{hc}{\lambda} = 0$

$$\Rightarrow \frac{-13.6eV \times 4}{n^2} + \frac{1224eVnm}{90nm} = 0$$

$\Rightarrow n = 2$

23. 9
Sol. According to stefan's law, the power radiated by a black body at absolute temperature T is given by

$\theta = \sigma AT^4$ …(1)

According to wein's displacement law

$\lambda_m T = b$

$\Rightarrow T = \frac{b}{\lambda_m}$ …(2)

From (1) and (2)

$$\theta = \sigma A\left(\frac{b}{\lambda_m}\right)^4$$

$$= \frac{\sigma A b^4}{\lambda_m{}^4}$$

For a sphere of radius r, $A = 4\pi r^2$

Hence $\theta = \frac{\sigma b^4 4\pi r^2}{\lambda_m^4} = K\frac{r^2}{\lambda_m^4}$

Where $K = 4\pi\sigma b^4$ is a constant.

Hence $\theta_1 = K\frac{4_1^2}{\left(\lambda_m^4\right)_1}$

$$\theta_2 = \frac{Kr_2^2}{(\lambda_m^4)_2}$$

$$\therefore \frac{\theta_1}{\theta_2} = \left(\frac{r_1}{r_2}\right)^2 \cdot \left(\lambda_m^4\right)_2$$

$$= \left(\frac{3}{5}\right)^2 \times \left(\frac{500}{300}\right)^4 = \left(\lambda_m^4\right)_1$$

$$= (5/3)^2$$

24. 15

Sol. $V_1 = \sqrt{\frac{2GM_E}{R_E}}$

$$\because \frac{1}{2}mV_2^2 - \frac{GM_E m}{R_E} - \frac{GM_S m}{R_S} = 0$$

$$\Rightarrow V_2 = \sqrt{\frac{2GM_E}{R_E}\left(1+\frac{k_1}{k_2}\right)}$$

$$\Rightarrow \frac{k_1}{k_2} = 15$$

25. 14

Sol. Let R be the radius of the sphere, therefore, potential on the surface is given by

$$V_0 = V_\infty + \frac{KQ}{R} = V_\infty + V$$

$$\Rightarrow V = \frac{KQ}{R}$$

$$\because V_P = V_\infty + \frac{KQ}{2R^3}\left(3R^2 - \frac{R^2}{4}\right) = V_\infty + \frac{11V}{8}$$

and $V_S = V_\infty + \frac{KQ}{2R} = V_\infty + \frac{V}{2}$

$$\Rightarrow V_P - V_S = \frac{7V}{8} = 14V$$

26. 10

Sol. formula, $\sum W = \Delta K$

$\because \frac{mg}{2}L - f_k L = 5J$ ……….. (1)

$-\frac{mg}{2}L - f_k L + FL = 5J$ …….(2)

$mgL - FL = 0$

$\Rightarrow F = mg = 10\text{ N}$

27. 11

Sol. formula, $\frac{dQ}{dt} = K\left(\overline{T} - T_S\right)$

$\because K\left(75 - T_S\right)t_1 = K\left(65 - T_S\right)t_2$

$\Rightarrow t_2 = 11\text{min}$

28. 98

Sol. formula, $f = \frac{n}{2L}V$

$\because \frac{f_0 + 5}{f_0 - 5} = \frac{1}{0.95}$

$\Rightarrow f_0 = 195\text{Hz}$

$\because f_0 = \frac{1}{2L}V \quad \Rightarrow L = \frac{V}{2f_0} = \frac{382.2}{2\times 195} = 0.98\text{m}$

29. 398

Sol. formula, polarisation $P = \sigma_{\text{induced}}$

$\because q = Q\left(1 - \frac{1}{K}\right)$

$\Rightarrow q = \frac{\varepsilon_0 A}{d}V\left(1 - \frac{1}{K}\right)$

$\Rightarrow \frac{q}{A} = \frac{\varepsilon_0 V}{d}\left(1 - \frac{1}{K}\right)$

$\Rightarrow \sigma_{ind} = 79.6\times 10^{-9}\,C/m^2$

30. 400

Sol. formula, $\frac{1}{f} = \left(\frac{\mu_\ell}{\mu_m} - 1\right)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$

$\frac{1}{f} = \frac{1}{400\text{cm}}$

$\frac{1}{f_{eq}} = \frac{2}{400\text{cm}} + \frac{2}{25\text{cm}} \quad \Rightarrow f_{eq} = \frac{200\text{cm}}{17}$

Therefore, distance is $2f_{eq} = \frac{400\text{cm}}{17}$

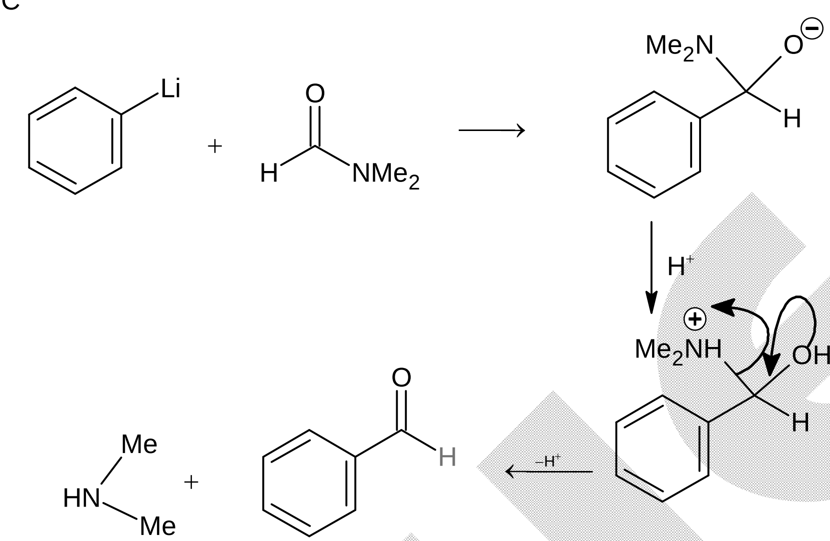
Chemistry

PART – B

SECTION – A

31. D
Sol. Vitamin A is fat soluble.

32. C
Sol.

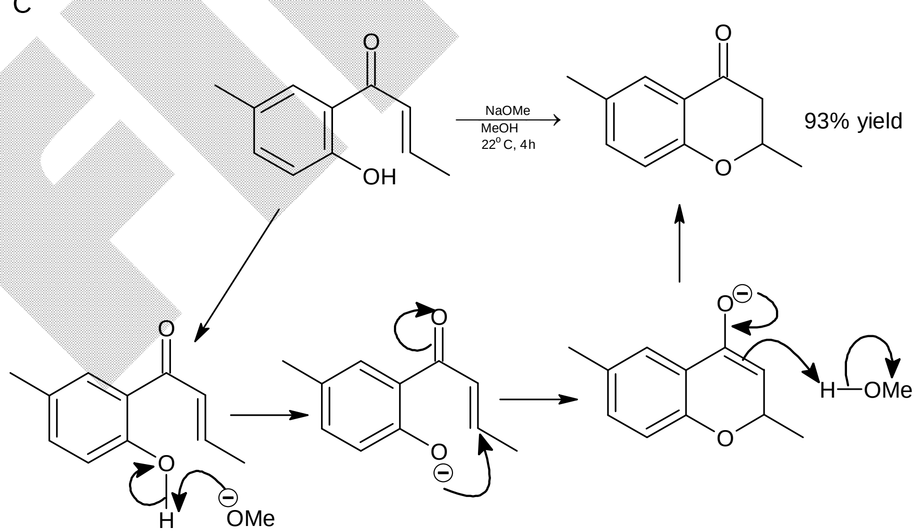


33. A
Sol. $T\ell$ has highest first ionization energy among the given elements.

34. B
Sol. $2KMnO_4 \longrightarrow K_2MnO_4 + MnO_2 + O_2$

35. A
Sol. The decomposition of N_2O on the Pt surface is a zero oder reaction.

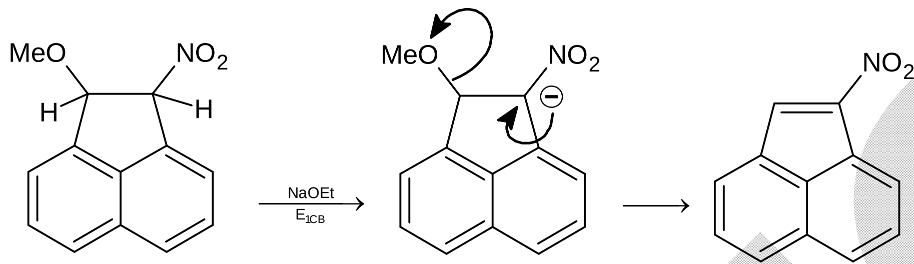
36. C
Sol.



37. C
Sol. N lone pair is not involved in back bonding.

38. D
Sol. As the electronegativity of central atom decreases bond angle decreases.

39. D
Sol.



40. B
Sol. Both are identical.

41. B
Sol. Due to d – d transition of Cr^{3+} ion in Al_2O_3 lattice.

42. C
Sol. $\left[Co(NH_3)_6\right]^{3+}$ has highest value of Δ_o.

43. C
Sol. As per law, each ion makes a definite contribution to total molar conductivity. Out of given solutions, only HCl in water is completely dissociated.

44. A
Sol. Moles $= \frac{1}{2} \times \frac{500}{1000} = \frac{250}{1000} \times 36.5 = 9.125$ g
Remaining weight of HCl = 9.125 – 1.335 = 7.79 g
$M = 0.569 = \frac{7.79}{36.5} \times \frac{1000}{V(ml)}$
$V = 375$ ml

45. C
Sol. $S^{2-} + \left[Fe(CN)_5(NO)\right]^{2-} \longrightarrow \underset{(Violet)}{\left[Fe(CN)_5 NOS\right]^{4-}}$

46. B
Sol. Nitrodealkylation.

47. B
Sol. The ΔH_{eg} of oxygen is lowest among group 16 elements.

48. D

Sol. $\frac{P^o - P}{P} = \frac{n_1}{n_2}$, $\frac{89.78 - 89}{89} = \frac{2/M}{100/78}$

M = 178

Number of C-atoms in each molecule $= \frac{178 \times \frac{94.4}{100}}{12} = 14$

Number of H-atoms in each molecule $= \frac{178 \times 5.6}{100} = 10$

49. D

Sol. Sodium amalgum is an alloy (mixture of sodium and mercury)

50. B

Sol. Ni^{2+} belongs to IV^{th} group of cations.

SECTION – B

51. 12

Sol. $\Delta U = 0$

$\Delta H = \Delta U + \Delta(PV) = 0 + B(P_2 - P_1)$

$-332 = B.\left(\frac{RT}{V_2 - B} - \frac{RT}{V_1 - B}\right)$

$V_1 = 12\ L$

52. 62

Sol. $\alpha = \frac{M_o - M}{(n-1).M} = \frac{208.5 - x}{(2-1)\times x} = 0.681$

x = 124

Equilibrium vapours are 62 times as heavy as hydrogen.

53. 136

Sol. Maximum solubility $4 \times 10^{-3}\ mol/L$

$= 4 \times 136 \times 10^{-3}\ g/L$

$= 0.544\ g/L$

$\therefore$ 0.544 g of $CaSO_4$ is present in 1 L.

$\therefore$ Initial volume of solution is 4 L, i.e. 4000 g solution.

$x = \frac{0.544}{4000} \times 10^6 = 136$ ppm

54. 390

Sol. Bond energy $= \left(\frac{hc}{\lambda} - \frac{2.5}{100}\left(\frac{hc}{\lambda}\right)\right) \times 6.0 \times 10^{23}$

$= \frac{97.5}{100}\left(\frac{hc}{\lambda}\right) \times 6.0 \times 10^{23}$

= 390 kJ/mol

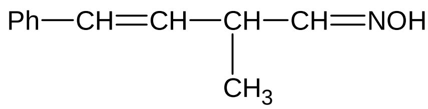
55. 8

Sol. Total number of geometrical isomers in $\left[Pt\left(H_2N - CH(CH_3)COO^-\right)_2\right]$ = 4

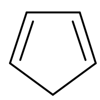
Total number of stereoisomer in $\left[Pt(gly)_3\right]^+$ = 4.

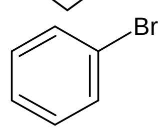
56. 6
Sol. Pt, Pd, Ru, Rh, Os, Ir are called platinum group elements.

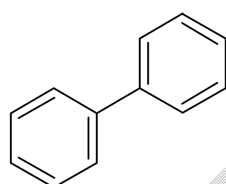
57. 11
Sol.



$$\frac{\text{number of stereoisomers + stereocentres}}{\text{number of chiral centres}} = 11$$

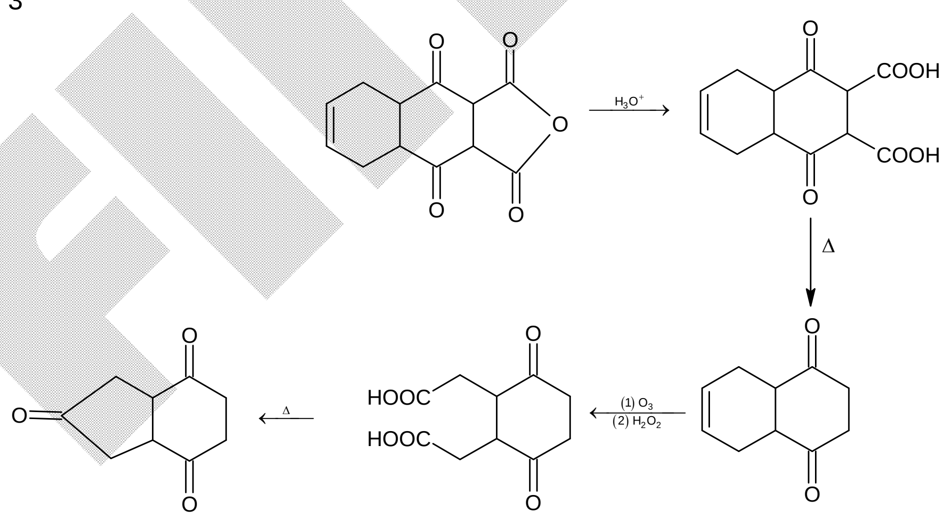
58. 15
Sol.
A is 

B is 

C is 

59. 4
Sol. Bond order of C_2 = 2
No. of $\pi-$bonds in C_2 = 2

60. 3
Sol.



Product

Number of keto groups in product = 3
The number of carbon atoms in product = 9.

Mathematics

PART – C

SECTION – A

61. B

Sol. $\left[{}^{1012}C_{1012}+{}^{1012}C_{1010}+{}^{1012}C_{1008}+........+{}^{1012}C_{0}\right]\times 2^{1012}-1=2^{2023}-1$

62. D

Sol. (x, y) lies on the angle bisector of $y-\sqrt{3}x=0$ and $\sqrt{3}y-x=0$

63. D

Sol. $p(x)=\dfrac{\sqrt{3}+|\tan x|}{1-\sqrt{3}|\tan x|}$

$x\in\left(\sqrt{3},3\right)\Rightarrow p(x)=\tan\left(\dfrac{\pi}{3}-x\right)$

64. B

Sol. $\lim\limits_{x\to 0^+}\left[\dfrac{a}{x}\right]\left[\dfrac{x}{c}\right]\ \lim\limits_{x\to 0^+}\left[\dfrac{b}{x}\right]\left[\dfrac{x}{d}\right]$

= 0 (since a, b, c, d > 0)

65. B

Sol. f(x) is discontinuous at x = $\left\{2n-\dfrac{1}{2},2n\right\}$ $\forall$ n $\in$ I

66. C

Sol. Given that $(\alpha\beta)^{2024}+(\gamma\delta)^{2024}=0\Rightarrow c=c'=0$

67. D

Sol. $2025^{2024}-2021^{2024}=(2023+2)^{2024}-(2023-2)^{2024}$

$=2[{}^{2024}C_1 2023^{2023}.2+{}^{2024}C_3 2023^{2021}2^3+\ldots\ldots]$

68. D

Sol. $\dfrac{\left(\sqrt{2}+\sqrt{2}\right)^{10}-\left(\sqrt{2}-\sqrt{2}\right)^{10}}{(2+1)^{10}+(2-1)^{10}}=\dfrac{2^{15}}{3^{10}+1}$

69. D

Sol. $0<1-\alpha<\dfrac{\pi}{2}\ \Rightarrow\ \alpha\in\left(1-\dfrac{\pi}{2},1\right)$

$0<\alpha^2-1<\dfrac{\pi}{2}\ \Rightarrow\ \alpha\in\left(-\sqrt{\dfrac{\pi}{2}+1},-1\right)\cup\left(1,\sqrt{\dfrac{\pi}{2}+1}\right)$

so $g(1-\alpha)$ and $g(\alpha^2-1)$ are not defined for any $\alpha\in R$

70. B

Sol. $f(x)\in\left(-1,-\dfrac{3}{4}\right]$

71. A

Sol. $\int_0^{\pi/2} x\sin x\,dx = \left|\frac{x^2}{2}\sin x\right|_0^{\pi/2} - \int_0^{\pi/2}\frac{x^2}{2}\cos x\,dx$

$= \frac{\pi^2}{8} - \left|\frac{x^3}{6}\cos x\right|_0^{\pi/2} - \int_0^{\pi/2}\frac{x^3}{6}\sin x\,dx$

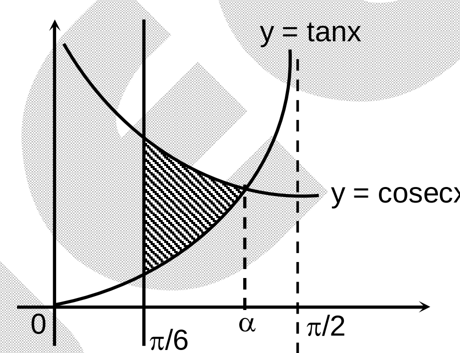
$= \frac{\pi^2}{8} - \frac{(\pi/2)^4}{24} + \int_0^{\pi/2}\frac{x^5}{4\cdot5\cdot6}\sin x\,dx$

72. B

Sol. cosec α = tanα

$\Rightarrow$ cosα = 1 – cos$^2\alpha$ = $\frac{-1+\sqrt{5}}{2}$

$A = \int_{\pi/6}^{\alpha}(\text{cosec}x - \tan x)dx$



$= \left|\ln\left|\frac{\text{cosec}x - \cot x}{\sec x}\right|\right|_{\pi/6}^{\alpha} = \left|\ln\left|\frac{\cos x - \cos^2 x}{\sin x}\right|\right|_{\pi/6}^{\alpha}$

$= \ln\left|\frac{2\cos\alpha - 1}{\sin\alpha}\right| - \ln\left|\frac{\frac{\sqrt{3}}{2} - \frac{3}{4}}{\frac{1}{2}}\right|$

$= \frac{1}{2}\ln\left(\frac{18-8\sqrt{5}}{\sqrt{5}-1}\right) - \ln\left(\frac{2\sqrt{3}-3}{2}\right)$

73. A

Sol. S : (x – 1)^2 + (y – 2)^2 + λ(x + y – 3) = 0

S passes through (3, 5) $\Rightarrow \lambda = -\frac{13}{5}$ centre of S : $\left(\frac{23}{10}, \frac{33}{10}\right)$

$\Rightarrow$ Equation of AC : $\frac{y-5}{x-3} = -\frac{7}{17}$

74. D

Sol. Equation of the hyperbola is xy – 5x – 8 = 0.

75. A

Sol. Locus of circumcentre is xy = 1.

76. C

Sol. Let A : (2λ + 1, λ + 3, 2λ + 2) and B : (μ + 2, 2μ + 2, 3μ + 3), then

$\overline{BA} : (2\lambda - \mu - 1)\hat{i} + (\lambda - 2\mu + 1)\hat{j} + (2\lambda - 3\mu - 1)\hat{k}$

$\Rightarrow$ 9λ – 10μ – 3 = 0 and 5λ – 7μ – 1 = 0 $\Rightarrow \mu = \frac{6}{13}$

So, B : $\left(\frac{32}{13}, \frac{38}{13}, \frac{57}{13}\right)$

77. A

Sol. Probability of any event cannot be greater than 1

78. B

Sol. Let $A(\vec{a}), B(\vec{b}), C(\vec{c})$ so $D\left(\frac{2\vec{b}+\vec{c}}{3}\right), E\left(\frac{4\vec{a}+\vec{b}}{5}\right)$ and $F\left(\frac{3\vec{c}+\vec{a}}{4}\right)$

Area(ΔDEF) = $\frac{1}{2}\cdot\frac{125}{300}\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|$

$\Rightarrow \frac{\text{Area}(\Delta ABC)}{\text{Area}(\Delta DEF)} = \frac{300}{125} = \frac{12}{5}$

79. A
Sol. Mode will increase by 10.

80. B
Sol. x = 2 is the only solution.

SECTION – B

81. 0
Sol. If z = x + iy then $x^2 + y^2 - x - y = 0$ and $x + y = 0 \Rightarrow x = 0, y = 0$

82. 768

Sol. $\sum_{A\in S} A = \begin{bmatrix} 256 & 256 & 256 \\ 256 & 256 & 256 \\ 256 & 256 & 256 \end{bmatrix}$

83. 1
Sol. For $\lambda = 1$, system of solutions is inconsistent.

84. 32
Sol. $2 + 3 \times 2 + 4 \times 6 = 32$

85. 135

Sol. $\frac{\frac{3x}{5}+\frac{3x}{5}+\frac{3x}{5}+\frac{3x}{5}+\frac{3x}{5}+y+\frac{z}{3}+\frac{z}{3}+\frac{z}{3}}{9} \ge \left(\frac{3^2}{5^5}(75)^7\right)^{1/9}$

$\Rightarrow 3x + y + z \ge 135$

86. 3

Sol. $\frac{dy}{dx} = \frac{y\ln y}{\ln x}\left[\frac{1}{x}+\frac{(x+1)\ln x}{x}+(\ln x)^2\right]$

87. 4

Sol. $8\pi r\frac{dr}{dt} = 4\pi r^2\frac{dr}{dt} \Rightarrow 2 = r$

88. 3
Sol. Let $\tan^{-1}(e^x) = \theta \Rightarrow e^x = \tan\theta$
$\Rightarrow \sin\theta\sec^2\theta\, d\theta = \sec^2\theta\, dy$
$\Rightarrow -\cos\theta = y + c \Rightarrow -\cos[\tan^{-1}(e^x)] = y + c$
$y(0) = 0 \Rightarrow c = -\frac{1}{\sqrt{2}}$

$x = \ln\sqrt{3} \Rightarrow -\frac{1}{2}+\frac{1}{\sqrt{2}} = y \Rightarrow y = \frac{\sqrt{2}-1}{2}$

89. 2

Sol. $\int\left[\left(x+\frac{1}{x}\right)\left(1-\frac{1}{x^2}\right)+\left(1-\frac{1}{x^2}\right)\right]e^{\left(x+\frac{1}{x}\right)}dx$

$= \left(x+\frac{1}{x}\right)e^{\left(x+\frac{1}{x}\right)}+c \Rightarrow g(1) = 2.$

90. 2

Sol. $\Delta_1 = 0$

$\Delta_2 = \frac{1}{2}2\sqrt{2}\left(4+\sqrt{2}\right) = 4\sqrt{2}+2$